The Ionosphere, Sky Wave, and AM Radio After Dark

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The ionosphere's layers

The ionosphere is the part of the upper atmosphere where sunlight has stripped electrons from atoms, leaving a thin plasma that can bend radio waves. It runs from about 37 mi (60 km) up to the top of the thermosphere, about 373 mi (600 km), and it changes shape between day and night.

Layer	Height	Daytime	Nighttime	Effect on radio
D	about 37–55 mi (60–90 km)	Present	Disappears within about an hour of sunset	Absorbs AM-band and lower shortwave signals by day
E	about 55–95 mi (90–150 km)	Strong	Weakens greatly	Reflects some signals, with shorter hops
F1	about 95–125 mi (150–200 km)	Present	Merges into F2	Reflects some shortwave
F2	about 125–250 mi (200–400 km), densest near 190 mi (300 km)	Strongest	Survives all night as a single F layer	Carries most long-distance shortwave

The lower layers fade at night because the air there is dense enough for electrons and ions to recombine quickly once sunlight stops. The F layer is so high and thin that recombination takes hours, so it lasts until sunrise.

How sky wave works

A sky wave is a radio signal that climbs into the ionosphere, bends back toward the ground, and lands far beyond the horizon. Shortwave listeners and ham radio operators call it skip.

- **Bending, not bouncing.** As the wave climbs into plasma with more and more free electrons, it travels a little faster there and curves gradually, like light bending in a lens. If the plasma is dense enough, the wave turns all the way around and heads back down.
- **Virtual height.** Seen from the ground, the wave acts as if it had hit a mirror at one height. Radio engineers call that the virtual height. For the F layer it is typically 150–250 mi.
- **Hop distance.** One hop off the F layer covers up to about 2,500 mi (4,000 km). A hop off the lower E layer covers up to about 1,200 mi (2,000 km).
- **Multiple hops.** The returning wave can reflect off the ground or ocean and climb again. Several hops can carry a signal around the world.
- **Skip zone.** Between the end of the ground wave and the first landing of the sky wave there is often a dead zone where the station can't be heard at all.

Why 3 to 30 MHz

3 to 30 MHz is the definition of the HF (high frequency) band, often called shortwave. It is also the range where sky wave works best around the clock. The two ends are set by different physics.

The top end: the wave escapes. How strongly the plasma bends a wave depends on its electron density, and higher frequencies need denser plasma to turn around. Above roughly 30 MHz, even the dense F2 layer can't bend the wave back, and it passes straight through to space. That is why FM radio, TV, and cell signals stay local, and why GPS and satellite signals get through to the ground. The highest frequency that still returns over a given path is called the maximum usable frequency. It rises in daytime, falls at night, and climbs much higher near the peak of the 11-year sunspot cycle.

The bottom end: the wave is absorbed. Lower frequencies reflect even more easily, but by day they must first pass through the D layer. The D layer sits low, where the air is dense enough that its electrons, set vibrating by the radio wave, keep colliding with air molecules and turn the wave's energy into heat. Lower frequencies lose more this way, so below about 3 MHz the daytime sky wave is mostly absorbed. After sunset the D layer disappears, and lower frequencies, including the AM band, skip just fine.

Even lower frequencies. Very long waves, below about 300 kHz, reflect off the bottom of the D layer itself and travel between it and the ground as if in a waveguide. Navies use them to reach submarines.

AM radio after dark

Many US AM stations must cut power, switch to a directional antenna, or go off the air at sunset, because the AM band starts to skip at night.

- **Daytime:** the D layer absorbs AM-band sky waves (530 to 1700 kHz). Stations reach listeners only by **ground wave**, which follows Earth's curved surface for roughly 50 to 150 mi, depending on power, frequency, and soil.
- **After sunset:** the D layer vanishes within about an hour. AM sky waves now reach the E and F layers and land hundreds to more than 1,000 mi away.
- **The problem:** many stations across the country share each AM frequency. At night their sky waves would land in each other's coverage areas and interfere.
- **The rules:** the FCC sorts AM stations into classes. A few **clear-channel (Class A)** stations keep full power at night, and their frequencies are protected across a wide region. Other stations on the same frequency must cut power, aim their signal away, or sign off at sunset. Daytime-only stations are Class D.
- **Chicago examples:** WGN 720 and WLS 890 are Class A clear-channel stations, heard across much of the country after dark.

A local case study: WAIT. Chicago's 820 kHz was home to the original WAIT, best known for its beautiful music format in the 1960s and 1970s. It signed off at sunset to protect WBAP in Fort Worth, Texas, the clear-channel station on 820 (source). In 1986 the 820 station changed its call letters, and WAIT moved to 850 kHz in Crystal Lake. That station also ran daytime only, signing off at sunset to protect KOA in Denver, the clear-channel station on 850 (source). Today 820 kHz in Chicago is WCPT, which uses a separate nighttime transmitter in Joliet (source).

Quick reference

Signal	Frequency	How it travels	Day	Night
Very low frequency (submarine radio)	below 300 kHz	Between the ground and the bottom of the D layer	Long range	Long range

Signal	Frequency	How it travels	Day	Night
AM broadcast	530–1700 kHz	Ground wave by day; sky wave off the E and F layers at night	Local only	Hundreds to 1,000+ mi
Shortwave (HF)	3–30 MHz	Sky wave off the F layer	Long range on higher frequencies	Long range on lower frequencies
FM, TV, cell, GPS	above about 30 MHz	Straight line; passes through the ionosphere	Line of sight	Line of sight

Check-for-understanding questions

1. Which ionosphere layer carries most long-distance shortwave, and about how high is it?
2. A sky wave doesn't really bounce. Describe what actually happens to it in the ionosphere.
3. Why can't FM radio signals skip off the ionosphere?
4. Why do low-frequency signals fail to skip during the day, and what changes at sunset?
5. Why does the FCC require many AM stations to reduce power or sign off at night?
6. Why did the old WAIT in Chicago have to sign off at sunset, while WGN 720 stays at full power all night?

Ionosphere heights and frequency limits are approximate and vary with time of day, season, and the sunspot cycle. Station history from WAIT (AM) and WCPT (AM), Wikipedia.

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